



WEEK 5 – SECURITY MODEL AND ACCESS CONTROL

End-to-End Salesforce CRM Transformation for Customer Operations and Service Management

5.1 Role Hierarchy Design

The role hierarchy for GreenGrid Utilities is designed based on the stakeholder influence assessment conducted during Milestone 1. The primary purpose of this structure is to provide controlled upward visibility of records while maintaining data security at lower organizational levels.

When Organization-Wide Defaults (OWD) are configured with restrictive access settings, the role hierarchy enables managers and executives to automatically access records owned by users below them in the reporting structure. This ensures that leadership teams can monitor operational performance without requiring manual sharing configurations.

To maintain consistency across the project deliverables, the Salesforce role structure follows the organizational groups identified during stakeholder analysis and process design activities.

Executive Management (Top-Level Role)

This role sits at the highest level of the hierarchy. Users assigned to this role require complete visibility into customer cases, utility assets, technician activities, and operational metrics to support strategic decision-making and executive reporting.

Customer Support Team (Operational Role)

Members of this role are responsible for handling incoming customer requests and creating service records. Their access is focused on customer service activities and does not automatically extend to field technician records.

Field Technicians (Operational Role)

Field Technicians operate alongside the Customer Support Team at the same organizational level. Users in this role primarily manage field service activities and maintain ownership of records associated with their assigned work.

5.2 Permission Set Definitions

Following the principle of least privilege, standard user profiles are configured with only the minimum permissions required for daily responsibilities. Additional capabilities are granted through Permission Sets when specific job functions require expanded access.

Since the data model relies heavily on Lookup Relationships, Permission Sets provide a flexible method for granting field-level and object-level permissions without modifying baseline profiles.

5.2.1 Field Technician Mobile Access

Permission Set Name:

Field_Technician_Mobile_Access

This Permission Set is assigned to users within the Field Technicians role to support mobile-based service operations and real-time data entry from field locations.

Object	Object Permissions	Field-Level Access
Account	Read Only	Access limited to customer identification and service location information.
Case	Read Only	Access limited to issue descriptions and case priority information.
Utility_Asset__c	Read Only	View access to utility asset identifiers and equipment information.
Field_Visit__c	Read, Create, Update	Full editing rights for Repair_Status__c and Technician_Notes__c fields.

5.2.2 Customer Support Escalation Access

Permission Set Name:

Support_Desk_Escalation_Access

This Permission Set is intended for senior support coordinators who require additional authority when managing critical service incidents or large-scale outage situations.

Object	Object Permissions	Field-Level Access
Case	Read, Edit, Delete	Allows ownership reassignment and modification of escalation-related fields.
Field_Visit__c	Read, Edit	Enables workload balancing and reassignment of technician activities when required.

5.3 Sharing Rules Configuration

To maintain data accuracy, prevent unauthorized access, and reduce the risk of duplicate information, record visibility is restricted by default. Sharing Rules are then applied selectively to grant access where business requirements justify broader visibility.

5.3.1 Organization-Wide Defaults (OWD)

The following Organization-Wide Default settings establish the baseline security model across the Salesforce environment.

Object Name	API Name	OWD Setting	Security Purpose
Account	Account	Private	Prevents unauthorized viewing of customer records across different operational regions.
Contact	Contact	Private	Protects customer contact details and communication history.
Case	Case	Private	Limits case visibility to authorized support personnel.

Utility Asset	Utility_Asset__c	Private	Secures utility infrastructure and equipment information.
Field Visit	Field_Visit__c	Private	Restricts technician work records to authorized users only.

5.3.2 Criteria-Based Sharing Rule

To support rapid decision-making during emergency situations, an automated sharing rule is implemented for critical service cases.

Rule Configuration

- Rule Name: **High_Priority_Case_Sharing**
- Object: **Case**
- Rule Type: **Criteria-Based Sharing**
- Criteria:
 - Priority = High
 - Status = New OR Working
- Shared With:
 - Executive Management Role
- Access Level:
 - Read/Write

Business Purpose

This rule automatically grants Executive Management visibility into critical incidents, ensuring leadership teams can monitor and respond to high-impact service disruptions without waiting for manual intervention.

5.4 Salesforce Implementation and Configuration

Step 1: Configure Organization-Wide Defaults

To establish the baseline security model:

1. Navigate to **Setup** → **Security** → **Sharing Settings**.
2. Select **Edit** within the Organization-Wide Defaults section.

3. Set the Default Internal Access value to **Private** for:
 - Account
 - Contact
 - Case
 - Utility_Asset__c
 - Field_Visit__c
4. Save the changes and wait for the recalculation process to complete.

The screenshot shows the Salesforce Sharing Settings page. The 'Organization-Wide Defaults' table is as follows:

Object	Default Internal Access	Default External Access	Grant Access Using Hierarchies
Lead	Public Read/Write/Transfer	Private	✓
Account and Contract	Private	Private	✓
Contact	Private	Private	✓
Order	Controlled by Parent	Controlled by Parent	✓
Asset	Controlled by Parent	Controlled by Parent	✓
Opportunity	Private	Private	✓
Field_Visit	Private	Private	✓
Utility_Asset	Private	Private	✓

Step 2: Configure the Role Hierarchy

To establish the organizational reporting structure:

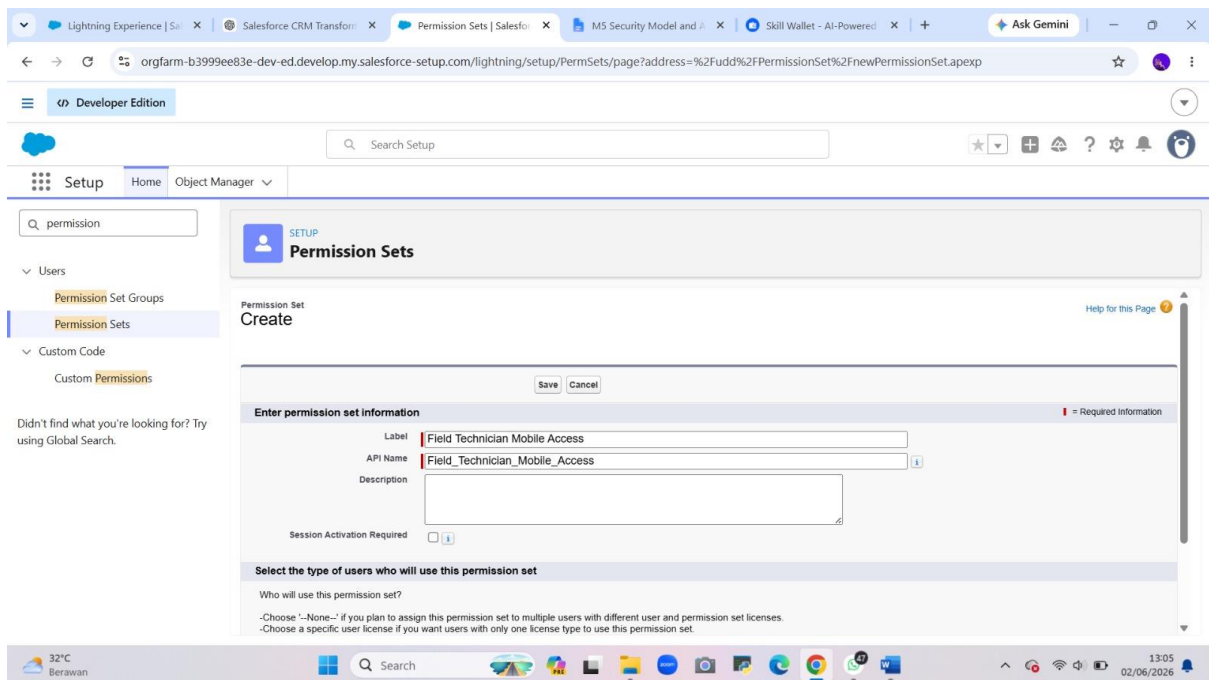
1. Navigate to **Setup** → **Users** → **Roles**.
2. Open the role hierarchy configuration page.
3. Edit the default CEO role.
4. Rename it as:
 - Role Label: **Executive Management**
 - Role Name: **Executive_Management**

5. Create a child role named **Customer Support Team**.
6. Create another child role named **Field Technicians**.
7. Save the hierarchy.

Step 3: Create the Field Technician Permission Set

To provide additional mobile-access permissions:

1. Navigate to **Setup** → **Users** → **Permission Sets**.
2. Select **New Permission Set**.
3. Create a Permission Set named:
 - **Field Technician Mobile Access**
4. Save the Permission Set.



Configure Object Permissions

1. Open **Object Settings**.
2. Select **Field Visit**.
3. Enable:
 - Read

- Create
 - Edit
4. Grant field editing access for:
- Repair Status
 - Technician Notes
5. Save the configuration.

The screenshot shows the Salesforce Setup interface for configuring a Permission Set. The left sidebar shows the navigation menu with 'Permission Sets' selected. The main content area is titled 'Permission Sets' and shows the 'Field Visits' configuration for a specific permission set. The 'Field Visits' section includes a 'Tab Settings' table and an 'Object Permissions' table. The 'Object Permissions' table shows that the 'Read' permission is enabled. The 'Field Permissions' table shows that the 'Repair Status' and 'Technician Notes' fields are selected for both 'Read Access' and 'Edit Access'.

Field Name	Field API Name	Read Access	Edit Access
Created By	CreatedById	☒	☐
Last Modified By	LastModifiedById	☒	☐
Owner	OwnerId	☒	☒
Parent Case	Parent_Case__c	☐	☐
Repair Status	Repair_Status__c	☒	☒
Resolution Timestamp	Resolution_Timestamp__c	☐	☐
Technician Notes	Technician_Notes__c	☒	☒
Visit Number	Name	☒	☐

Step 4: Configure the Automated Sharing Rule

To automatically expose high-priority cases to management:

1. Navigate to the **Case Sharing Rules** section.
2. Create a new sharing rule.
3. Configure:
 - Rule Name: **High Priority Case Sharing**

- Rule Type: **Criteria-Based**
- 4. Set Criteria:
 - Priority = High
- 5. Share records with:
 - Executive Management Role
- 6. Assign:
 - Read/Write Case Access
- 7. Save the rule.

The screenshot shows the Salesforce Setup interface for configuring a sharing rule. The left sidebar shows the navigation menu with 'Setup' selected. The main content area is titled 'Sharing Settings' and displays the following steps:

- Step 1: Rule Name**
 - Label: High Priority Case Sharing
 - Rule Name: High_Priority_Case_Sharing
 - Description: (empty)
- Step 2: Select your rule type**
 - Rule Type: ☒ Based on record owner ☐ Based on criteria
- Step 3: Select which records to be shared**

Criteria	Field	Operator	Value	
	Priority	equals	High	AND
	--None--	--None--		AND
	--None--	--None--		AND
	--None--	--None--		AND
	--None--	--None--		AND

Additional Options: ☒ Include records owned by users who can't have an assigned role
- Step 4: Select the users to share with**
 - Share with: Roles | Executive Management
- Step 5: Select the level of access for the users**
 - Case Access: Read/Write

Buttons: Save, Cancel

Step 5: Create Test Users and Assign Roles

To validate the security configuration, create representative users for each operational role.

Customer Support User

- First Name: Support
- Last Name: Agent

- Username Example: support123@greengrid.dev
- Role: Customer Support Team
- User License: Salesforce
- Profile: Standard User

Save the user record and create the next user.

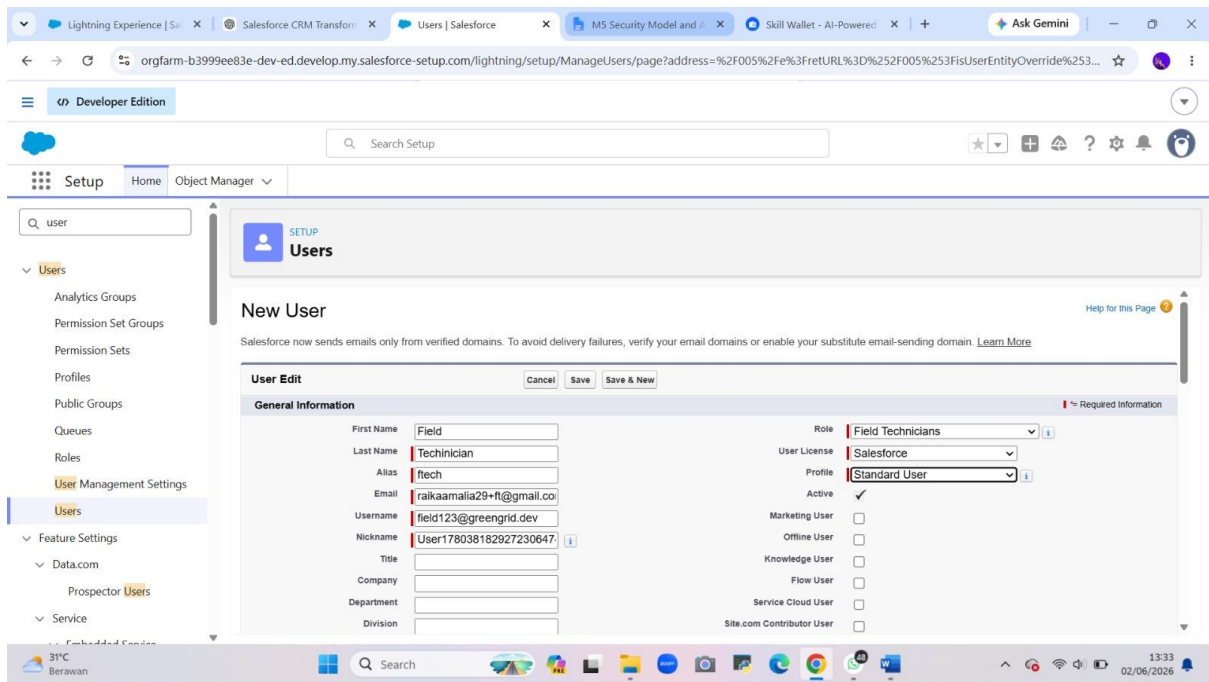
The screenshot shows the Salesforce 'New User' setup page. The 'User Edit' form is displayed with the following details:

- General Information:**
 - First Name: Support
 - Last Name: Agent
 - Alias: sagen
 - Email: raikaamalia29+cs@gmail.co
 - Username: support123@greengrid.dev
 - Nickname: User178038149886832498
 - Title:
 - Company:
 - Department:
 - Division:
- Role and License:**
 - Role: Customer Support Team
 - User License: Salesforce
 - Profile: Standard User
- Active and Other Users:**
 - Active: ☒
 - Marketing User: ☐
 - Offline User: ☐
 - Knowledge User: ☐
 - Flow User: ☐
 - Service Cloud User: ☐
 - Site.com Contributor User: ☐

Field Technician User

- First Name: Field
- Last Name: Technician
- Username Example: field123@greengrid.dev
- Role: Field Technicians
- User License: Salesforce
- Profile: Standard User

Save the user record.



Validation

Return to the Active Users page and confirm that both users:

- Are active
- Have the correct assigned roles
- Have the correct profiles
- Appear within the appropriate hierarchy positions

Conclusion

This milestone establishes a comprehensive Salesforce security framework that balances data protection with operational accessibility. By combining Role Hierarchies, Permission Sets, Organization-Wide Defaults, and automated Sharing Rules, GreenGrid Utilities can ensure that users have access only to the information necessary for their responsibilities while maintaining executive visibility into critical business operations. This security architecture also provides a scalable foundation for future growth and governance within the Salesforce CRM platform.